

# Vishal Kumar Mathuri

SOLIDITY / ETHEREUM BLOCKCHAIN DEVELOPER | SMART CONTRACTS / DEFI / WEB3

📞 9082135308 ✉ vishal789mathur@gmail.com 📍 Patna, India 🔗 [linkedin.com/in/vishal-mathuri/](https://www.linkedin.com/in/vishal-mathuri/) 📄 [github.com/vishalmathuri](https://github.com/vishalmathuri) 🌐 Website: [vishal-website-beta.vercel.app](https://vishal-website-beta.vercel.app)

## SUMMARY

Blockchain Developer specializing in **Solidity and the Ethereum ecosystem**, with hands-on experience building smart contracts, DeFi protocols, NFT platforms, access-control systems, and full-stack Web3 applications — including **50+ smart contracts** and **10+ full-stack blockchain applications** shipped to date. Strong focus on **secure and testable smart-contract engineering** using Solidity, Foundry, Hardhat, OpenZeppelin, and Ethers.js, with full-stack experience across React.js, Node.js, and REST APIs.

Experienced across **DeFi lending and liquidation, NFTs, decentralized identity, and role-based access control**, including non-traditional Web3 integrations such as Telegram-based gaming — with a portfolio centered on production-oriented, security-conscious protocol design.

## TECHNICAL SKILLS

**Smart Contracts:** Solidity, ERC-20, ERC-721, access control, staking, governance, vesting, custom errors, events  
**EVM / Blockchain:** Ethereum, EVM, Polygon, Ethereum Sepolia, gas optimization, contract verification, IPFS  
**Security / Testing:** OpenZeppelin, ReentrancyGuard, AccessControl, Ownable, Pausable, unit testing, fuzz testing, invariant testing  
**Web3 / Backend:** Ethers.js, Web3.js, React.js, Node.js, Express.js, REST APIs, MetaMask  
**Tools:** Foundry, Hardhat, Remix, Git, GitHub, VS Code, Vercel, Render, IPFS/Pinata

## PROFESSIONAL EXPERIENCE

**Blockchain Developer** Jul 2025 - Present  
Freelance

- Developed and deployed **50+ Solidity smart contracts** implementing ERC-20, ERC-721, staking, governance, and marketplace functionality across NFT, fundraising, identity, and access-control applications.
- Built **10+ full-stack Web3 applications** integrating smart contracts with **React.js frontends, Node.js backend services, REST APIs, wallets, and EVM networks**.
- Applied **OpenZeppelin security primitives** and secure design patterns including reentrancy protection and role-based access control, reducing security risk across deployed contracts.
- Worked across contract implementation, testing, deployment, frontend integration, debugging, and production issue resolution.

**Senior Content Developer** Jul 2019 - Mar 2023  
Think & Learn Pvt Ltd Bengaluru, IN

- Managed scheduling operations for 200+ student batches across multiple grade levels with **100% operational accuracy** and enhanced workflow efficiency.
- Improved an automated concept-tagging workflow to **95% classification accuracy** through systematic validation and quality checks, boosting system efficiency and reducing errors.
- Crafted and executed structured Learning Journeys for academic programs utilizing digital learning platforms, resulting in a **96% success rate**.
- Designed and curated interactive educational content, mock tests, and learning modules for Byju's TV and BHELP programs, increasing user satisfaction and engagement.
- Resolved student support and academic operations requests using Salesforce with **98% resolution accuracy** within 24 hours, ensuring uninterrupted course delivery.
- Collaborated with academic operations, HR, and support teams while maintaining strong cross-team communication and coordination.

## FLAGSHIP BLOCKCHAIN PROJECTS

### 1. Production-Grade EVM Lending Protocol

**Tech Stack:** Solidity • Foundry • Chainlink • OpenZeppelin

**Links:** [GitHub](#)

- Engineered a modular, overcollateralized EVM lending protocol supporting multi-asset supply, withdrawal, borrowing, repayment, interest accrual, and liquidation, with responsibilities separated across **LendingPool, ReserveManager, RiskManager**, and configurable interest-rate modules.
- Implemented **index-based scaled accounting and risk management**, including utilization-driven borrow/supply rates, collateral-factor borrowing limits, health-factor solvency checks, liquidation thresholds/bonuses, multi-decimal asset normalization, and rounding-aware debt/collateral accounting.
- Built a **Chainlink-based oracle layer** with defensive price validation, including stale-price protection, incomplete/stale round detection, positive-price validation, future-timestamp checks, configurable feed heartbeats, and an oracle abstraction enabling deterministic mock testing.
- Validated the protocol with an extensive **Foundry test suite spanning unit, integration, fuzz, and stateful invariant testing**, plus failure-path and security suites; deployed and source-verified the contracts on Ethereum Sepolia and executed an end-to-end supply → borrow → repay → withdraw flow.

### 2. NFT Marketplace

**Tech Stack:** Solidity • ERC-721 • OpenZeppelin • React.js • Ethers.js • IPFS/Pinata

**Links:** [GitHub](#) • [Live \(Vercel\)](#) • [API \(Render\)](#)

- Engineered a full-stack NFT marketplace supporting ERC-721 minting, listing, purchasing, and ownership transfers through Solidity smart contracts, with a React.js interface for end-to-end marketplace interactions.
- Integrated **Ethers.js and MetaMask** for wallet-based Web3 workflows, enabling users to connect accounts, sign transactions, mint NFTs, list assets, and execute purchases directly against deployed smart contracts.
- Implemented decentralized NFT metadata storage using **IPFS/Pinata**, linking token metadata and digital assets to ERC-721 tokens while using OpenZeppelin contracts for standardized NFT functionality and ownership controls.

- Delivered a publicly accessible full-stack application, deploying the React frontend on Vercel and backend services on Render to provide a live, testable Web3 application beyond standalone smart contracts.

3. Scai Warrior — Web3 Telegram Game

Tech Stack: Solidity • Ethereum • JavaScript • Ethers.js • Telegram • Web3

Links: [GitHub](#) • [Live Demo](#)

- Built a Web3-enabled Telegram gaming application combining a Telegram bot, Solidity smart contracts, backend infrastructure, and browser-based game components into a single cohesive product.
- Developed the Telegram bot layer to handle player onboarding, game session management, and in-game actions, bridging conversational bot interactions with on-chain gameplay logic.
- Integrated Ethers.js and wallet-based Web3 workflows to connect blockchain transactions and game state directly to Telegram users without requiring a separate app install.
- Demonstrated blockchain integration inside a non-traditional application environment, combining Telegram, wallets, and smart contracts into a working, publicly deployed game.

4. Soulbound Identity System

Tech Stack: Solidity • ERC-721 • OpenZeppelin • Hardhat • Ethers.js • Ethereum

Links: [GitHub](#)

- Designed a decentralized identity system built on non-transferable Soulbound Tokens (SBTs) to represent permanent, verifiable blockchain-based credentials.
- Implemented ERC-721-based token contracts with transfer restrictions enforced at the contract level, ensuring credentials remain permanently bound to their original recipient.
- Built credential issuance and verification logic using OpenZeppelin standards, enabling authorized issuers to mint and manage identity tokens securely.
- Demonstrated a practical approach to on-chain identity and credentialing, illustrating non-transferable token design patterns relevant to reputation and verification systems.

INTERNSHIPS

Blockchain Developer

EtherAuthority

Mar 2026 - May 2026

Remote

- Built a Telegram Mini App featuring multiple in-app mini-games, including Coinflip, Find the King (a dice-based game), and Wheel, using the Telegram Web App platform.
- Designed and implemented game logic, betting flow, and result computation for each game mode.
- Collaborated with the team on frontend/backend integration and testing to ship engaging, low-latency gameplay experiences.

Blockchain Developer

Leafny Web Pack

May 2025 - Jul 2025

Remote

- Assisted in building Web3 dApps integrating smart contracts with frontend and backend systems.
- Worked with ERC-20 and ERC-721 token standards and participated in smart-contract security testing and reviews.

Blockchain Developer

Estatos

Jan 2025 - Feb 2025

Remote

- Developed and optimized smart contracts for blockchain operations.
- Worked on tokenization and DeFi protocols, contributing to secure and efficient smart-contract deployments.
- Collaborated with the tech team to enhance solutions while demonstrating problem-solving and a growth mindset.

Solidity Developer

Nexus Network

Jul 2024 - Dec 2024

Remote

- Developed bridge contracts for relayer-based cross-chain transactions.
- Strengthened smart-contract security and optimization practices.
- Actively participated in team collaborations and implemented feedback for continuous improvement.

EDUCATION

Post Graduate Diploma in Blockchain

IIIT Bangalore & upGrad

Mar 2023 - Mar 2024

Bengaluru, IN

- Completed advanced coursework covering JavaScript, Node.js, Ethereum smart-contract development, and Hyperledger Fabric.
- Developed Ethereum-based smart-contract architectures for decentralized applications including DeFi systems.
- Designed system architecture for the UPSTAC healthcare application, focusing on scalable and secure blockchain integration.
- Explored blockchain-based KYC systems, Hyperledger Fabric network setup, and cross-chain frameworks.

Dual Degree (B.Tech + M.Tech), Energy Science & Engineering

Indian Institute of Technology Bombay (IIT Bombay)

Jul 2014 - Jul 2019

Mumbai, IN

- Secured 61%.

ADDITIONAL INFORMATION

- **Languages:** Fluent in Hindi (Native) and English (Professional)
- **Interests:** Reading and Cycling